ZCAS UNIVERSITY

SCHOOL OF COMPUTING AND APPLIED SCIENCES

BACHELOR OF INFORMATION TECHNOLOGY/BACHELOR OF NETWORK ENGINEERING

CIT3242 IT PROJECT MANAGEMENT

END OF SEMESTER FINAL EXAMINATION

WEDNESDAY 4TH DECEMBER 2024

09:00 – 12:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS: READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This section is compulsory and must be attended.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination carries a total of **100 marks.**
4. Candidates must not turn this page until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

QUESTION 1.

Read the following and then answer the questions that follow.

Imagine you are a project manager tasked with overseeing the development of a new software application for a leading technology company. The project involves a cross-functional team of developers, designers, and quality assurance professionals. As you kick off the project, you need to ensure effective project cost management to stay within budget and deliver a high-quality product on time.

Required:

a) Explain how you would determine the initial budget for the software development project, considering the various cost elements such as personnel, technology, and other resources.

(8 Marks)

b) What methodologies or techniques would you employ to estimate the costs associated with the development, testing, and deployment phases of the software project?

(8 Marks)

c) How do you account for unforeseen circumstances or changes in project scope that may impact the budget? **(8 Marks)**

d) What contingency planning measures would you put in place to handle unexpected cost increases?

(8 Marks)

e) Describe the mechanisms and tools you would use to monitor project costs throughout its lifecycle.

(8 Marks)

[Total: 40 Marks]

15 + 10 + 5 + 8

10
10
8

43 1

43/100 × 60/1 = x

43
6 1

25.8

25.8
15.2 1

SECTION B: Attempt any THREE questions in this section.

QUESTION 2.

You are a project manager who has completed the development of a new mobile application for a client. The project involved designing, developing, testing, and launching the app. You have met all the project objectives and completed it on time and within budget. The mobile app is now available to users, and your client is satisfied with the results.

Required:

a) Explain the significance of the project completion and evaluation phase in the project management life cycle. What are the key activities and deliverables involved in this phase when concluding the development of the mobile app in your scenario? **(5 Marks)**

b) In your mobile app project, you have implemented a post-implementation review (PIR) process. Describe the purpose of a PIR and provide examples of the questions or criteria you would use to assess the project's success and identify areas for improvement. **(5 Marks)**

c) You have been asked to conduct a financial analysis of your mobile app project. Describe the financial metrics and calculations you would use to evaluate the project's performance, including return on investment (ROI), cost-benefit analysis, and payback period. **(5 Marks)**

d) In your mobile app project, you have identified lessons learned and best practices. Discuss the importance of documenting lessons learned and how this information can be used to improve future projects. Provide examples of lessons learned and how they might be applied to future endeavours. **(5 Marks)**

Total: 20 Marks]

QUESTION 3.

You are the project manager for a software development project that aims to create a new student database management (SDM) system for ZCAS University. The SDM system is expected to streamline record keeping, fast information retrieval and improve overall business efficiency for the university. As the project manager, you understand the critical importance of project quality management to ensure that the delivered SDM system meets or exceeds client expectations and industry standards.

Required:

a) Explain how you would approach the planning phase of project quality management for the SDM system development. What specific quality standards or methodologies would you consider, and how would you ensure they align with the client's expectations? **(6 Marks)**

b) Describe the steps you would take to implement a robust quality assurance process throughout the software development lifecycle. How would you ensure that quality is not only monitored but also proactively built into the project from the early stages? **(6 Marks)**

c) In the testing phase of the SDM system, what specific quality control measures would you implement to identify and address defects or issues? How would you prioritize and manage the resolution of identified quality issues? **(4 Marks)**

d) How do you plan to involve the client in the quality management process? What strategies would you employ to gather feedback, ensure alignment with client expectations, and make any necessary adjustments to the project deliverables? **(4 Marks)**

[Total: 20 Marks]

QUESTION 4.

You are a project manager responsible for implementing a new customer relationship management (CRM) system for a medium-sized company. The project aims to improve customer interactions, streamline sales processes, and enhance data management. You are in the early stages of project implementation.

Required:

a) Explain the significance of the project implementation phase in the project management life cycle. Discuss key activities and considerations you need to address during this phase in the context of implementing the CRM system for the company. **(5 Marks)**

b) You would like to make use of a Breakdown structure in this CRM system. Explain what the breakdown structure is and provide the advantages of the breakdown structure. **(5 Marks)**

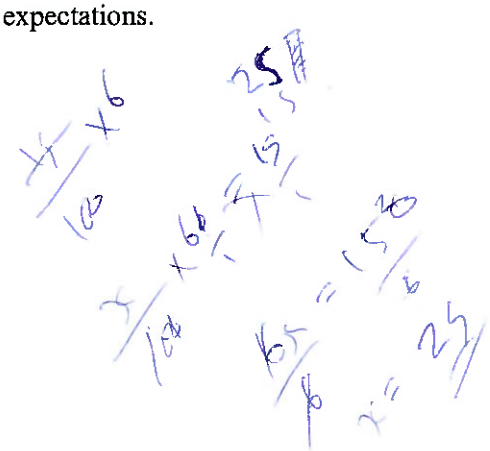
c) To implement the CRM system, you need to be a good leader. Provide 5 steps of good project leaders. **(5 Marks)**

d) What are project Milestones? Using an example explain how you could develop milestones for the CRM project. **(5 Marks)**

[Total: 20 Marks]

QUESTION 5.

You are leading a project team responsible for the construction of a new office building for a multinational corporation. The project involves multiple stakeholders, including architects, construction contractors, project sponsors, and various regulatory authorities. Effective communication is critical to ensure that the project progresses smoothly, stays on schedule, and meets the client's expectations.



Required:

a) Explain how you would prioritize and engage with different stakeholders to ensure effective communication throughout the various phases of the office building construction project.

(5 Marks)

b) Describe the steps you would take in developing a communication plan for the project. How does the plan address the diverse needs and preferences of the project team members and stakeholders?

(5 Marks)

c) In the event of unexpected changes or disruptions to the project timeline or scope, how would you communicate these changes to the project team, sponsors, and other stakeholders?

(5 Marks)

d) Explain how you would foster a culture of open communication and collaboration within the project team? What tools and techniques would you use to ensure that team members are consistently sharing information and updates? **(5 Marks)**

[Total: 20 Marks]

END OF FINAL SEMESTER EXAMINATION